

Postpartum Hemorrhage 08: Delayed PPH, Anemia, Recovery, and Final Synthesis

Companion reading handout for simultaneous listening.

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Delayed PPH, anemia, recovery, and final synthesis.

The bleeding has stopped.

The uterus is firm.

The chart says stable.

The team has left the room.

But the patient is exhausted, pale, afraid to stand, trying to breastfeed, replaying what just happened, and wondering whether bleeding will start again.

PPH management is not finished when the blood stops.

That is the theme of the final episode.

The patient still has a postpartum course.

We will cover delayed postpartum hemorrhage, suspicious placenta follow-up, postpartum anemia, recovery, thrombosis prevention, and then close the course with a compact algorithm.

Let us begin with delayed PPH

Primary PPH is within the first 24 hours.

Secondary, or delayed, PPH occurs after 24 hours and up to 12 weeks postpartum.

Bleeding that requires medical attention occurs in about 1 to 2 percent of postpartum women.

A common timing is days 8 to 14.

Why then?

One reason is placental site healing.

As the placental site eschar sloughs, bleeding can transiently increase.

Not every increase in bleeding is an emergency.

But persistent heavy bleeding deserves evaluation.

A useful practical question is:

Does the bleeding settle within 1 to 2 hours, or does it continue?

If it continues, if it is heavy, if there are symptoms, fever, pain, dizziness, weakness, foul discharge, or hemodynamic changes, evaluate.

Delayed PPH is a diagnostic mystery with a short list of suspects.

Is this normal eschar slough?

Is there retained tissue?

Is there infection?

Is there subinvolution of the placental site?

Is there coagulopathy?

Is there von Willebrand disease?

The evaluation starts as usual, with the patient.

Vital signs.

Hemodynamic assessment.

Pelvic examination.

Uterine size and tenderness.

Bleeding amount.

Fever or infection signs.

Complete blood count.

Coagulation tests when clinically indicated.

Ultrasound can help distinguish an empty cavity, clot, and retained tissue.

But ultrasound is an adjunct, not an oracle.

Clot and retained fragments can be difficult to distinguish.

The suspicious placenta protocol gives an important warning: immediate postpartum ultrasound is not a reliable way to diagnose retained remnants, and should be used at the discretion of a senior physician.

The same caution applies more broadly.

Do not use ultrasound alone as an automatic rule-out.

Do not use ultrasound alone as an automatic curettage trigger.

Interpret it with the clinical picture.

Bleeding more than 48 hours postpartum should make you think about von Willebrand disease.

Why?

Von Willebrand factor rises in pregnancy.

After delivery, levels can fall, and bleeding may be delayed.

So delayed bleeding after 48 hours is a signal to consider screening, especially if the history supports it.

Now treatment

If the uterus is empty and atony or subinvolution is suspected, uterotonics may help.

If there is significant retained material or heavy bleeding, suction evacuation or curettage may be needed.

But curettage in the postpartum uterus is not routine scraping.

The postpartum uterine wall is soft and can perforate easily.

The suspicious placenta protocol also emphasizes that postpartum curettage is associated with intrauterine adhesions, with an approximate 40 percent figure cited in that protocol, and may impair fertility.

Only about one third of postpartum women who underwent curettage in the departmental experience had placental tissue found.

So the lesson is balanced:

Curettage can save the patient when retained tissue or heavy bleeding requires it.

Careless or unnecessary curettage can harm her.

Use it when indicated, carefully, and with senior judgment.

When curettage is performed, broad-spectrum antibiotics with anaerobic coverage are recommended before the procedure, especially if a sharp curette is used.

If bleeding is refractory to uterotonics or curettage and the patient is stable enough, embolization can be considered.

Now suspicious placenta follow-up

The local protocol requires physician examination of every suspicious placenta.

The mother should be monitored carefully during hospitalization regardless of the placental examination and regardless of whether a scan was performed.

She should be told what complications may occur and what treatments might be needed.

At discharge, she should be instructed that heavier-than-usual bleeding or fever can occur and that she must return for problems during the postpartum period.

These instructions must be documented.

Follow-up should be arranged with a senior physician when indicated by the protocol.

This may sound administrative.

It is not.

It is delayed hemorrhage prevention.

Good discharge counseling is a clinical intervention.

Now postpartum monitoring after the acute event

After vaginal birth, the postpartum protocol includes blood pressure, pulse, temperature, saturation, uterine fundal height and contraction, vaginal bleeding, perineum, pain, epidural sensation, urination and mobility, breastfeeding, and mental state.

The nurse repeats the physical assessment two hours after arrival to the ward and again four hours later.

The doctor must be notified for blood pressure below 90 over 50, pulse above 105 or below 60, difficulty breathing or saturation below 94 percent, faintness, dizziness, weakness, increased vaginal bleeding, uncontracted uterus, no urination within 8 hours of delivery or catheter removal, or abnormal pain.

When a doctor is called, a complete blood count should be performed with the clinical assessment.

After vaginal delivery, complete blood count is performed on the first postpartum morning, or earlier if symptoms or abnormal findings require it.

It should be repeated for hemoglobin below 8, hemoglobin below 9 with symptoms, hemoglobin drop greater than 3 grams per deciliter from baseline, unstable hemoglobin after two or more tests, white blood count above 20,000, or platelets below 100,000.

After cesarean delivery, complete blood count is performed 6 hours after surgery and again on postoperative day one.

Monitoring is frequent: every hour for the first 4 hours, every 2 hours from 4 to 12 hours, every 4 hours from 12 to 24 hours, and then every 12 hours from the second

day onward.

Report triggers after cesarean include low or high blood pressure, pulse above 105 or below 60 in repeated tests, breathing difficulty, faintness or weakness, distended abdomen, uncontracted uterus, increased vaginal bleeding or wound bleeding, bloody urine, urine output less than 35 milliliters per hour over the last two hours, severe pain despite analgesia, or patient or family request.

That last one matters.

The patient or family request is a report trigger.

They may see deterioration before the chart does.

Now anemia

The puerperium position paper defines postpartum anemia as hemoglobin below 10 grams per deciliter.

Postpartum anemia is common. Developed-country studies report 10 to 30 percent, and a local study found 12 to 19 percent.

The main causes are iron deficiency and significant delivery blood loss.

Anemia is not only a number.

It is fatigue.

Reduced cognitive ability.

Impaired quality of life.

Increased risk of postpartum depression.

Difficulty caring for the newborn.

Difficulty breastfeeding.

Difficulty standing without fear.

So recovery after PPH includes treating anemia, not merely documenting it.

Continue iron or multivitamin supplementation for at least 6 weeks postpartum.

Continue folic acid throughout the reproductive period.

Include complete blood count in postpartum follow-up and continue treatment according to results.

Now breastfeeding and mother-infant recovery

Breast milk is recommended as optimal infant nutrition, and exclusive breastfeeding for six months is recommended in the position paper.

But after PPH, breastfeeding may be affected by fatigue, anemia, pain, separation, intensive monitoring, surgery, emotional distress, and delayed lactogenesis.

The resident should not treat this as secondary.

Ask about feeding.

Ask about dizziness while holding the baby.

Ask about pain.

Ask whether she can sit up.

Ask whether she feels safe standing.

Ask whether she understands what happened.

The event may be over medically, but not psychologically.

Recovery is part of hemorrhage care.

Now venous thromboembolism prevention

The postpartum period is a high-risk time for thrombosis, and cesarean delivery increases risk.

But immediately after PPH, anticoagulation must be balanced against bleeding control.

The local thrombosis prevention table says that when shock or dehydration occurred during delivery and the shock stems from PPH, prophylactic treatment can be initiated approximately 24 hours after delivery.

That does not mean automatic treatment at 24 hours in every case.

It means consider prophylaxis when bleeding is controlled and the clinical context allows it.

Balance ongoing bleeding, surgical hemostasis, epidural catheter timing, platelet count, anesthesia rules, and the patient's thrombosis risk.

Heparin and warfarin are not contraindicated during breastfeeding according to the local table.

Now let us close the whole course

PPH is a race between blood loss and source control.

The uterus is the tourniquet.

Pregnancy hides shock.

Visual estimation underestimates bleeding.

Hemoglobin is late.

The first response is parallel.

One hand on the uterus.

Two large lines.

Three calls.

Four T's.

One person at the uterus.

One person at the veins.

One person at the phone.

One person thinking.

Oxytocin first.

Tranexamic acid early.

Choose the next uterotonic by contraindication.

Hypertension or preeclampsia means no Methergine.

Asthma or active pulmonary disease means no carboprost.

Misoprostol is micrograms, never milligrams.

Firm uterus, keep looking.

Tone is the soft uterus.

Trauma is bleeding despite tone.

Tissue is the incomplete or trapped placenta.

Thrombin is the patient who cannot clot.

Bakri is a test, not a delay.

Stable enough for embolization.

Unstable enough for surgery.

Hysterectomy saves lives.

Red cells carry oxygen.

Factors make clot.

Fibrinogen matters early.

Calcium, potassium, temperature, and acidosis are part of hemorrhage care.

Accreta is planned hemorrhage.

And the patient still has a postpartum course.

Here is the final one-minute verbal algorithm.

Call for help.

Quantify bleeding.

Massage the uterus and empty the bladder.

Place two large-bore lines.

Send complete blood count, type and crossmatch, coagulation profile, and fibrinogen.

Notify blood bank, senior obstetrician, anesthesia, laboratory, and operating room.

Give oxytocin and tranexamic acid.

Choose additional uterotonics by contraindication.

Put your hand on the uterus.

If soft, treat tone.

If firm, inspect for trauma, tissue, and thrombin.

Repair tears.

Remove retained tissue when indicated.

If bleeding continues after two uterotonics or crosses escalation thresholds, move to Bakri, operating room, embolization, surgery, and massive transfusion according to stability.

If coagulopathy appears, replace blood products and clotting factors while source control continues.

If accreta is suspected, stop pulling and call help.

After the bleeding stops, monitor, treat anemia, counsel return precautions, support breastfeeding and recovery, and reassess thrombosis prevention.

That is the course.

Postpartum hemorrhage is not a single maneuver.

It is recognition, choreography, medication safety, cause-specific diagnosis, procedural escalation, transfusion physiology, planned prevention, and postpartum recovery.

The resident who manages it well is not the one who memorizes the longest list.

It is the one who keeps asking the right question at the right time:

Where is the blood coming from, what is the patient becoming physiologically, and what must happen next to stop both?